

each file access permission is represented by a number. The numbers are as follows:-

- -r (read): 4
- -w (write): 2
- -x (execute): 1
- - (no permission): 0

You can make different combinations of these values to make different combinations of permissions given to the particular user for a particular file. The combinations are given below:-

NUMBER	PERMISSION
-	-
1	-x
2	-w-
3 (1+2)	-wx
4	-r
5	r-x
6	rw-
7	rwX

Now let us take an example and see how file permissions are changed in real time

Q. Suppose you have a file named `chris.txt` with permissions `rwXrwr--`, you need to change the permission so that everyone can read and execute.

Ans. `chmod 555 chris.txt`

Applying chmod in Symbolic mode to change the permissions

The major drawback with Absolute mode was that you need to always provide three numbers for changing permissions even when you need to change permission for just one owner. This is where symbolic mode will come into play because here owners are denoted by the following symbols and you don't need to mention all of them every single time.

In symbolic mode, owners are denoted by the following symbols:-

- u = user owner
- g = group owner
- o = other
- a = all which means user + group + other